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# RAG-msCNN: predicting protein-DNA binding sites by integration of retrieval-augmented generation (RAG) with protein language models and the multi-scale separable convolutional neural network

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ABSTRACT

*Background and objective:* Protein function and molecular regulatory mechanisms require the precise identification of nucleic acid-binding residues. Although some more recent deep learning algorithms have advanced, it is still challenging to extract contextual information specific to residues from sequences.
*Methods:* In this paper, we propose a novel deep learning architecture called RAG-msCNN that improves sequence representation by combining Retrieval-Augmented Generation (RAG) with Multi-Scale Separable Convolutional Neural Networks (msCNN). Our approach shows better generalization and robustness on benchmark DNA- and RNA-binding datasets even without structural input.
*Results:* Our method improves performance on several benchmark datasets, such as TE46, TE181, and TE129, by combining protein language models with multi-scale feature learning and residue-specific RAG integration. Our framework is better than other methods, with an AUC of up to 99.9 % and an F1 score of more than 91 %. This shows that it can accurately and reliably find metal-binding sites.
*Conclusions:* The framework is a scalable and robust solution for predicting residue-level interactions, and it generalizes well to both DNA and RNA interaction tasks. This work provides a strong and comprehensible tool for nucleic acid-binding prediction and protein functional annotation in situations with limited data or structured and unbalanced datasets.

## 1. Introduction

The interactions between proteins and nucleic acids, particularly DNA and RNA, are crucial for cellular processes such as transcription, replication, translation, and repair [1–3]. During these interactions, genes are expressed when and where they are needed, thereby influencing cellular fate and function. Various diseases, such as cancer, are associated with disruptions in protein-nucaleic acid interactions, which emphasizes the need to identify and characterize these interacting sites accurately to diagnose and treat them [4]. It has been demonstrated that mutations in DNA-binding proteins are associated with abnormal gene expression and contribute to the development of cancer [5]. Identifying the specific residues involved in molecular interactions between nucleic acid-binding proteins is critical to the development of targeted therapeutic interventions. Additionally, it is imperative to understand how RNA-binding proteins interact with their target RNAs to develop therapies for viral infections in which viral RNA interferes with host proteins to hijack cellular functions [6].

Transcriptional factors, polymerase, and repair enzymes rely on hydrogen bonds and electrostatic attraction to interact with DNA [7,8]. There are several RNA-binding proteins (RBPs) that regulate mRNA splicing, stability, and translation through interactions with mRNA, rRNA, tRNA, and non-coding RNAs [2,9]. Accurate identification of binding sites facilitates gene regulation mechanisms and targeted therapies development [1]. X-ray crystallography, electrophoretic mobility shifting analysis (EMSA), nuclear magnetic resonance (NMR), and cryo-

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electron microscopy (cryo-EM) are conventional methods that provide highly precise identification of interacting residues. Although these methods are time-consuming, expensive, and impractical for high-throughput analyses [10–12]. Recent advances in machine learning, particularly deep learning, have demonstrated their effectiveness at predicting binding sites. It is possible to predict nucleic acid-binding residues using a variety of computational methods, such as sequence-based methods that rely on physicochemical properties and structure-dependent methods that depend on protein structure. Some sequence-related algorithms, including TargetDNA [13], DNAPred [14], and RNABindRPlus [15], use machine learning algorithms to identify features associated with protein sequences. Some methods utilize structural features, such as aaRNA [16], NucleicNet [17], and DNABind [18], including deep learning algorithms, such as convolutional neural networks, to improve accuracy. Recently, graph-based methods, such as GraphBind [19] and other GNNs [20–22], have proved promising for capturing spatial relationships and learning complex structural patterns related to binding. However, these methods are still unable to capture the nuanced interactions between sequence, structure, and evolutionary information that are crucial to accurately predicting binding sites.

Even though these methods have improved significantly in terms of prediction accuracy and generalizability, they still need to be improved. Existing methods have difficulty distinguishing binding residues from non-binding residues, especially when subtle structural or physicochemical changes are involved. To enhance the performance and robustness of these prediction tools, it is necessary to develop algorithms that integrate multi-scale information derived from protein sequences, structures, and evolutionary histories. Recently, several pre-trained protein language models (PLMs) may be useful for predicting interacting sites, such as ProtTrans, ESM-1b, and ESM-2 [23–25]. By using large databases of protein sequences, these models can capture context information and evolutionary relationships within proteins. Through the combination of embeddings and deep learning architecture, more accurate and robust predictions can be achieved by capturing the complex interactions between sequence, structure, and evolution.

This study addresses the problem of highly unbalanced datasets encountered in biological problems, such as binding site prediction, where interacting residues are substantially fewer than non-interacting residues. We utilized the Retrieval-Augmented Generation (RAG) strategy, a natural language model framework that incorporates dense retrieval and generation [26], to address this issue specifically. Unlike pre-trained PLM embeddings, which capture global contextual representations, the RAG module enhances these embeddings by collecting similar sequences or residue fragments from a reference database. This retrieval adds functional and evolutionary information relevant to the target residue, broadening the feature space and increasing discriminative power for locating rare binding residues.

Recent advances in convolutional neural networks (CNNs) indicate a greater emphasis on interpretability and efficiency. The EA-CNN [27] used attention and explainable AI (XAI) to accurately classify fruits and vegetables, whereas a lightweight CNN [28] using fractional optimization and XAI enhanced malaria detection convergence and interpretability. Dai et al. created a precise DNA methylation typing framework that uses hierarchical clustering of nanopore current signals and attention-based neural networks. This framework improves the resolution of epigenetic analysis [29]. Other studies have provided systematic comparisons and statistical frameworks for the detection of genomic islands and the identification of genome boundaries from one or more genomes [30–32]. Deep learning–based sequencing quality control and recursive feature selection approaches have also been employed to improve protein structural class prediction and DNA modification identification [33,34]. Our previous work had used mCNN architecture [35–39] to extract local patterns from protein sequences. The model used multiple parallel convolutional layers varying in kernel width to capture features across residues. Convolutional layers were concatenated and passed through fully connected layers that were then pooled and flattened. Although this model performed well, it was more expensive to train. Additionally, the model did not exploit channel-wise independence across feature dimensions, which can be valuable in high-dimensional embeddings like protein language embeddings. To overcome these limitations, we proposed a new model called Multi-Scale Separable Convolutional Neural Network (msCNN), which replaces standard convolutions with depth-wise separable convolutions. Decoupling spatial and channel-wise feature extraction reduces the number of trainable parameters and improves model efficiency across various datasets.

This study presents a novel deep learning method known as RAG-msCNN that uses pre-trained PLM embeddings and multi-scale features derived from protein structures in combination with a retrieval-augmented generation (RAG) method combined with the deep learning method msCNN. RAG-msCNN aims to enhance learning by retrieving relevant information from external databases, such as DNA-binding motifs and structural templates, and incorporating msCNN to improve the model's ability to generalize to unknown data. To overcome the challenges associated with highly imbalanced datasets, this integrated approach provides a more accurate and robust solution to DNA-binding site prediction.

## 2. Materials and methods

Fig. 1 illustrates the comprehensive research workflow of the current study. This includes data collection, feature extraction, multi-scale feature preparation, RAG adoption, and deep learning networks to achieve the objective of our study, which is enhancing DAN-binding site prediction.

### 2.1. Data collection

To predict protein-DNA interactions, we collected two datasets previously evaluated and established by previous researchers, Dataset 1 [40] and GraphBind Dataset 2 [19]. According to Table 1, we used the same methods of data collection and data processing as the original authors, including DNA-binding residue criteria and data integrity measures. In Dataset 1, TR646 consists of 646 proteins containing 314,139 residues, of which 15,636 are DNA-interacting sites, while the remaining 298,503 are non-interacting sites. There are 46 proteins in the test set TE46, which comprise 10,876 residues in total, of which 965 residues are binding and 9,911 residues are non-binding. GraphBind provides a training set, TR573, with 573 proteins, as well as two testing sets, TE181 and TE129, which contain 181 and 129 proteins, respectively. Training set TR573 includes 159,883 residues, including 14,479 binding sites and 145,404 non-interacting residues, while test set TE181 contains 75,258 residues, of which 3,208 are binding sites and 72,050 are non-interacting, and TE129 has 37,515 residues, of which 2,240 are labeled as binding and 35,275 as non-binding.

A CD-HIT algorithm [41] was used to ensure that no proteins in the training and test sets shared more than 30 % similarity to prevent data overlap between the two sets. Additionally, prior research revealed significant class imbalances in both datasets, with 4.98 % of DNA-binding residues constituting the majority of TR646 data compared to 9.06 % in TR573. These datasets are analyzed using identical class characteristics, keeping consistency with the original distribution, which poses a challenge to model training. By utilizing these previously curated datasets, we can compare our proposed approach to well-known data and ensure comparability with other protein-DNA binding discoveries.

In addition, as shown in Table 2, we used a benchmark RNA-binding protein dataset from PDNAPred [42] to evaluate our model's generalizability and robustness. Our DNA experiments used the same 30 % sequence identity cutoff for training and testing RNA datasets. In the RNA dataset, there is a training subset (TR495) containing 495 proteins and a testing subset (TE117) including 117 proteins. TR495 consists of

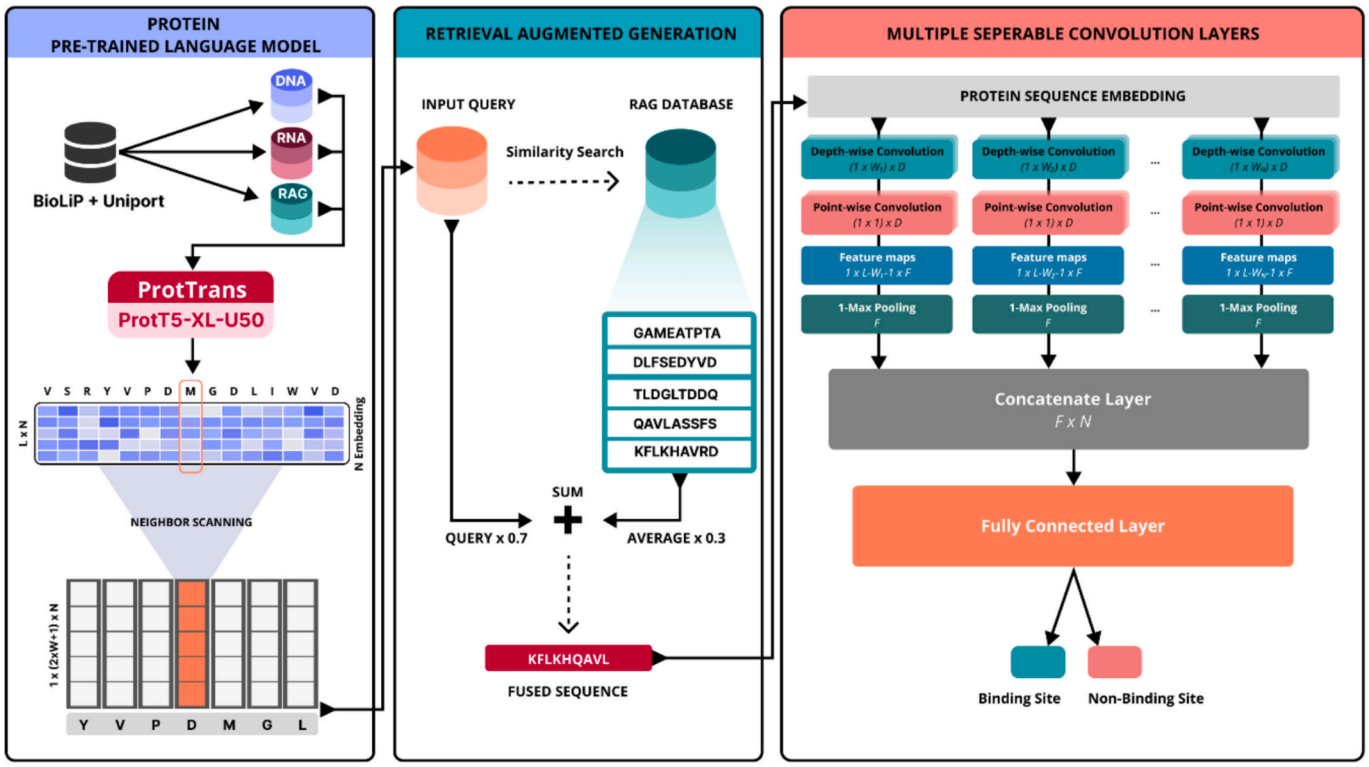



**Fig. 1.** Workflow of the DNA-binding site prediction architecture, utilizing pre-trained language models along with the RAG strategy and multi-scale separable convolutional neural network.

**Table 1**
Data distribution summary for benchmark protein-DNA binding datasets.

| | Dataset 1 | | Dataset 2 | | |
|---|---|---|---|---|---|
| **Dataset** | **TR646** | **TE46** | **TR573** | **TE181** | **TE129** |
| Number of proteins | 646 | 46 | 573 | 181 | 129 |
| Number of sites | 314,139 | 10,876 | 159,883 | 75,258 | 37,515 |
| Number of binding sites | 15,636 | 965 | 14,479 | 3,208 | 2,240 |
| Number of non-binding sites | 298,503 | 9,911 | 145,404 | 72,050 | 35,275 |
| % of binding sites | 4.98 % | 8.87 % | 9.06 % | 4.26 % | 5.97 % |

**Table 2**
Summary of benchmark protein–RNA binding data distributions.

| **Dataset** | **TR495** | **TE117** |
|---|---|---|
| Number of proteins | 495 | 117 |
| Number of sites | 136,899 | 37,345 |
| Number of binding sites | 14,609 | 2,031 |
| Number of non-binding sites | 122,290 | 35,314 |
| % of binding sites | 10.76 % | 5.44 % |

136,899 residues in total, of which 14,609 are binding residues and 122,290 are non-binding residues. TE117 consists of 37,345 residues, including 2,031 binding residues and 35,314 non-binding residues. In addition to our nucleic acid-interacting protein model, this additional RNA dataset lets us demonstrate its adaptability to a variety of nucleic acid-interacting proteins.

### 2.2. Feature extraction

#### 2.2.1. ProtTrans model for protein languages

In bioinformatics, language models are essential for analyzing complex data within protein sequences. The ProtTrans [23] model uses self-supervised learning to identify intrinsic patterns in data without requiring explicit labels. ProtTrans was developed using a variety of protein sequences primarily from the UniProt database. Various self-supervised learning activities enable the model to gain a comprehensive understanding of protein sequences, and after training, embeddings for native protein sequences are obtained. The embeddings were produced using a ProtTrans model with 24 layers and over 1.2 billion parameters. By converting protein sequences into ProtTrans embeddings, we were able to acquire the necessary contextual information and sequence data. The ProtTrans method is effective in distinguishing functional changes in proteins, emphasizing the context-specific effects of DNA/RNA binding sites in a protein. Using embeddings with 1024 dimensions, the dataset is uniform in size and representation.

#### 2.2.2. Feature representation

We adopted a multiscale residue-level feature representation strategy to capture both the local context and the global semantics of each protein. In protein sequences, we constructed contextual windows of size $2w + 1$, where $w = 7$, encompassing the target residue and its neighboring residues-seven on the left, seven on the right. The model can learn informative features based on short- and medium-range interaction between residues, which are essential for identifying functional binding sites.

$S = \{s_1, s_2, s_3, \cdots, s_n\}$ represent a protein sequence of length n, where each $s_i$ is an amino acid residue. For each residue $s_i$, we define a context window $W_i$ as follows:

$$W_i = \{s_{i-w}, s_{i-(w-1)}, \cdots, s_{i-1}, s_i, s_{i+1}, \cdots, s_{i+(w-1)}, s_{i+w}\} \quad (1)$$

The zero-padding method assigns a padding token to out-of-bound positions (i.e., residues close to the sequence start or end). As a result, each residue is assigned to a fixed-size context window.

Each residue $s_j \in W_i$ is transformed into a dense vector $v_j \in R^d$ using a protein language model (e.g., ProtT5). We can obtain the contextual representation $C_i$ for the center residue by concatenating the embedding

vectors of all residues in the window:

$$C_i = \{v_{i-w}, v_{i-(w-1)}, \cdots, v_{i-1}, v_i, v_{i+1}, \cdots, v_{i+(w-1)}, v_{i+w}\} \quad (2)$$

The model thus receives a feature representation for each residue of shape $(2w + 1) \times d(2w + 1) \times d$, where $d = 1024$ is the embedding size. Each residue is represented in its local environment, with its structural and contextual richness important for downstream tasks such as identifying binding sites.

#### 2.2.3. Retrieval-Augmented Generation (RAG) strategy

The external database (RAG-DB) was built by collecting protein sequences from a curated RNA-binding protein dataset. To avoid potential label leakage and data redundancy, CD-HIT was utilized to cluster all protein sequences at a 30 % sequence identity level [41]. This guaranteed that the sequences used for RAG retrieval did not overlap with those used for model training or testing, resulting in rigorous data independence and avoiding unwanted bias. With a sliding context window around each residue and embeddings generated by a pre-trained protein language model (e.g., ProtT5), the RAG-DB provides distinct contextual samples. This study improves on our previous RAG-based frameworks by incorporating a more context-aware retrieval fusion process that incorporates L2-regularized similarity estimation and adaptive weighting (70 % intrinsic, 30 % retrieved embeddings). We apply our RAG approach exclusively to positive residues and enhance their embeddings by using their vectors (v + ) as queries. By using Euclidean distance (L2), we identify the top k = 5 nearest neighbors (smallest distances) from database embeddings (u). By retrieving the top five similar binding residue embeddings from our database, we compute their average as:

$$c = \frac{1}{5}\sum_{j=1}^{5} v_j^+ \quad (3)$$

To fuse the contextual vector with the original embedding, 70 % of the original features are retained while 30 % of the retrieval patterns are incorporated to get the final embeddings as follows:

$$c = \alpha v^+ + (1 - \alpha)c \quad (4)$$

In this equation, we set α = 0.7 to ensure that the augmented embedding is within the convex hull of the original and retrieved embeddings. The result is a method that enhances the features of positive samples without diluting the context of the original data.

### 2.3. msCNN model architecture

Our proposed model, a Multi-Scale Separable Convolutional Neural Network (msCNN), analyzes high-dimensional protein sequence embeddings, as shown in Fig. 1. Input to the model is a 2D tensor with shape (1, MAXSEQ, NUM_FEATURE), where MAXSEQ is the fixed sequence length and NUM_FEATURE is the size of the embedding per residue. To reduce computational costs and improve learning efficiency, msCNN replaces standard convolution layers with SeparableConv2D layers. Each convolution is assigned a kernel width from a set of predetermined window sizes. Under separable convolution, the model can learn channel-specific relationships more efficiently while capturing the localized patterns across sequences. Following a MaxPooling2D operation that compresses the temporal dimension, the outputs of each parallel convolution path are flattened and concatenated to create a single high-dimensional feature vector. To prevent overfitting, a dropout layer of 0.7 % is applied before feeding the features into a fully connected layer activated by ReLU. Final classification is done by a dense layer regularized with an L2 penalty (λ = 1e − 3). Through separable convolution, this model learns multi-scale contextual features from protein embeddings, improving generalization and computational efficiency and eliminating manual bias of initialization.

## 3. Performance evaluation

This study examines the prediction capability of protein language models (PLMs), as well as our proposed RAG-ProtTrans for detecting DNA and RNA binding sites. The proposed architecture was tested using three different test sets, enabling a comprehensive examination of our architecture and validation. A sensitivity analysis was conducted to determine how the model responds to binding interactions. Specificity analysis ensured accuracy in identifying true non-binders through the model's ability to eliminate non-binding sites. According to the model's precision, it is effective in identifying true binding predictions among all positive predictions. With accuracy, a comprehensive performance metric was provided across all classifications, whereas with MCC, a balanced perspective was provided by integrating all four classification results. In addition, we assessed the model's performance using the F1 score, which provides a balanced measure of precision and recall, effectively capturing its ability to identify DNA-binding residues while minimizing false positives and false negatives, and AUC provided a threshold-independent assessment. These measures provide detailed insight into the model's ability to detect DNA and RNA binding sites, improving our understanding of DNA interactions. To calculate sensitivity, specificity, accuracy, MMC, and F1 scores, equations (5) through (9) are used.

$$Sensitivity, Recall = \frac{TP}{TP + FN} \quad (5)$$

$$Specificity = \frac{TN}{TN + FP} \quad (6)$$

$$Accuracy = \frac{TP + TN}{(TP + FP + TN + FN)} \quad (7)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (8)$$

$$F1 - Score = \frac{2TP}{2TP + FP + FN} \quad (9)$$

TP indicates the correct positive identification, TN the correct negative identification, FP the incorrect positive identification, and FN the incorrect negative identification.

## 4. Results

We evaluated our proposed model by experimenting with various window sizes, window combinations, and numbers of convolutional filters to determine its effectiveness and generalizability. To select the best configuration, we used three independent benchmark datasets: TE181, TE129, and TE46.

### 4.1. Comprehensive evaluation of DNA binding site identification with different test sets

Results in Table 3 provide a comprehensive evaluation of the performance of the given metrics values for DNA-binding site prediction across three distinct test sets. Our results indicate that our model consistently performs well across a variety of datasets. We observed a

**Table 3**
Performance evaluation of DNA binding site identification with different test sets.

| Dataset | Spe | Rec | Pre | F1 | MCC | AUC |
|---|---|---|---|---|---|---|
| TE181 | 0.9918 | 0.9934 | 0.8434 | 0.9123 | 0.9114 | 0.9998 |
| TE129 | 0.9938 | 0.9955 | 0.9109 | 0.9514 | 0.9492 | 0.9999 |
| TE46 | 0.9980 | 0.9523 | 0.9768 | 0.9217 | 0.9163 | 0.9896 |

specificity (Spe) of 0.9918 and a recall (Rec) of 0.9934 on the TE181 dataset and a precision (Pre) of 0.8434, giving an F1 score of 0.9123, a Matthews Correlation Coefficient of 0.9114, and an AUC of 0.9998, indicating excellent discrimination capability. In TE129, similar results were observed, including a precision of 0.9109, an F1 score of 0.9514, an MCC of 0.9492, and an AUC of 0.9909. Our results demonstrate that our strategy is robust and effective, particularly when it comes to addressing DNA binding site imbalances. In the third dataset, TE46, a specificity of 0.9980 demonstrated excellent non-interacting residue identification. The model achieved a recall of 0.8725 and a precision of 0.9768, resulting in a strong F1 score of 0.9217. In an extremely imbalanced test scenario, MCC of 0.9163 and AUC of 0.9896 confirm the model's effectiveness in handling sparse positive examples. Overall, our proposed model for DNA binding site prediction showed robustness, precision, and generalizability across all three test sets, despite variations in data size, binding site proportion, and sequence diversity.

### *4.2. The RAG ablation study and protein language model contribution analysis*

We conducted an ablation study using three different datasets to rigorously evaluate the effectiveness of our proposed RAG-ProtTrans strategy. To assess the improvement introduced by the retrieval and fusion mechanism, we will quantify the contribution of RAG-based augmentation when combined with pre-trained protein language model embeddings. Table 4 shows how each component impacts final model performance across multiple evaluation metrics derived from three different test sets. On the TE181 benchmark dataset, sequence-based embeddings such as ProtTrans, ESM-2 (1280), ESM-2 (480), ProtT5, ESM-C, and TAPE showed modest predictive performance, with F1 scores ranging from 0.1543 (TAPE) to 0.2623 (ProtTrans). Additionally, the ProstT5 structural embedding model failed to significantly improve predictions, with an F1 score of 0.2353 and an MCC of 0.2726. We also tested embeddings obtained from ESM-C, a contrastive variant of the ESM family, and the performance of ESM-C was notably lower than that of ProtTrans. A combined RAG strategy with ProtTrans embeddings, on the other hand, significantly improved predictive power, achieving F1 scores of 0.9123, MCCs of 0.8123, and AUCs of 0.998, which indicates a clear distinction between DNA-binding residues and non-binding residues. RAG-based embeddings outperform conventional and structure-aware embeddings in DNA-binding site prediction because of context-aware retrieval and sequence embeddings. Fig. 2 illustrates the ROC curves for these comparisons.

As a benchmark, TE129 was used, and baseline models with pre-trained embeddings failed, with F1 scores between 0.2396 and 0.3644 and MCCs below 0.3884. Although advanced models like ESM-2 (1280), ESM-C, ESM-2 (480), and ProstT5 captured more sequence and structural features, they achieved only modest improvements, with F1 scores between 0.2931 and 0.3619 and MCCs below 0.39. Our RAG-ProtTrans model, on the other hand, achieved a 0.9514 F1-score, 0.9188 MCC, and 0.999 AUC, with both specificity and recall near perfect. Aside from reinforcing the advantage of retrieval-augmented contextual learning, these findings also confirm that the RAG-based strategy can be generalized across multiple datasets, particularly DNA-binding site prediction, a domain with extreme class imbalance. An ROC-AUC comparison is shown in Fig. 3.

For the TE46 test set, the RAG-ProtTrans strategy demonstrated a clear performance advantage over all other embedding methods. In terms of specificity and recall, the RAG-ProtTrans strategy scored the highest across all metrics with a specificity of 0.9980, a recall of 0.9523,

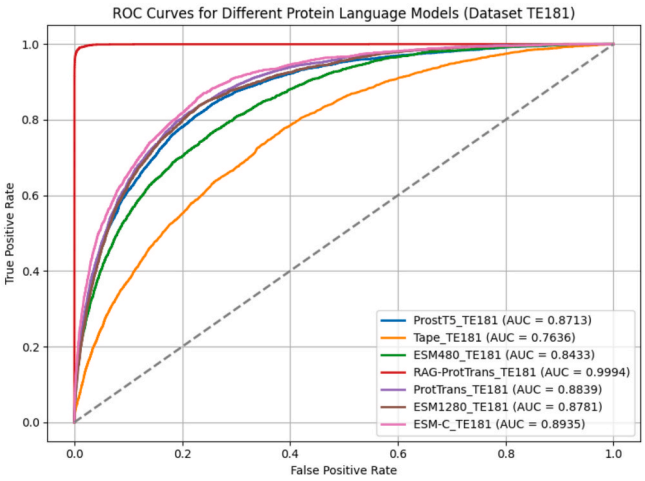



**Fig. 2.** ROC-AUC comparisons of RAG-ProtTrans with different pre-trained language model embeddings on the test set (TE181).

**Table 4**
Evaluation of independent test set performance using various embeddings of language models and RAG-ProtTrans embeddings.

| Dataset | Feature sets | Spe | Rec | Pre | F1 | MCC | AUC |
|---|---|---|---|---|---|---|---|
| TE181 | Prottrans | 0.8020 | 0.8263 | 0.1559 | 0.2623 | 0.3033 | 0.894 |
| | ESM-2 (1280) | 0.7684 | 0.8373 | 0.1379 | 0.2368 | 0.2791 | 0.8781 |
| | ESM-2 (480) | 0.7414 | 0.7694 | 0.1164 | 0.2021 | 0.2291 | 0.8433 |
| | ProstT5 | 0.7753 | 0.8102 | 0.1376 | 0.2353 | 0.2726 | 0.8713 |
| | Tape | 0.6600 | 0.7260 | 0.0864 | 0.1543 | 0.1624 | 0.76 |
| | ESM-C | 0.7706 | 0.7892 | 0.1322 | 0.2264 | 0.2594 | 0.8646 |
| | **RAG-Prottrans** | **0.9799** | **0.9855** | **0.8434** | **0.9123** | **0.8123** | **0.998** |
| TE129 | Prottrans | 0.8158 | 0.8688 | 0.2305 | 0.3644 | 0.3884 | 0.917 |
| | ESM-2 (1280) | 0.8268 | 0.8237 | 0.2319 | 0.3619 | 0.3770 | 0.9010 |
| | ESM-2 (480) | 0.7636 | 0.8112 | 0.1789 | 0.2931 | 0.3065 | 0.8638 |
| | ProstT5 | 0.7991 | 0.8321 | 0.2083 | 0.3332 | 0.3509 | 0.8950 |
| | Tape | 0.7241 | 0.7272 | 0.1434 | 0.2396 | 0.2328 | 0.8051 |
| | ESM-C | 0.7721 | 0.8576 | 0.1929 | 0.3149 | 0.3379 | 0.8935 |
| | **RAG-Prottrans** | **0.9903** | **0.9857** | **0.9109** | **0.9514** | **0.9188** | **0.999** |
| TE46 | Prottrans | 0.9815 | 0.2808 | 0.5969 | 0.3820 | 0.3730 | 0.8783 |
| | ESM-2 (1280) | 0.9718 | 0.3596 | 0.5543 | 0.4362 | 0.4046 | 0.8670 |
| | ESM-2 (480) | 0.9891 | 0.1803 | 0.6170 | 0.2791 | 0.3031 | 0.8245 |
| | ProstT5 | 0.7682 | 0.8166 | 0.2554 | 0.3891 | 0.3689 | 0.8694 |
| | Tape | 0.7043 | 0.7202 | 0.1917 | 0.3028 | 0.2560 | 0.7858 |
| | ESM-C | 0.7924 | 0.7617 | 0.2632 | 0.3912 | 0.3606 | 0.8519 |
| | **RAG-Prottrans** | **0.9980** | **0.9523** | **0.9768** | **0.9217** | **0.9163** | **0.9896** |

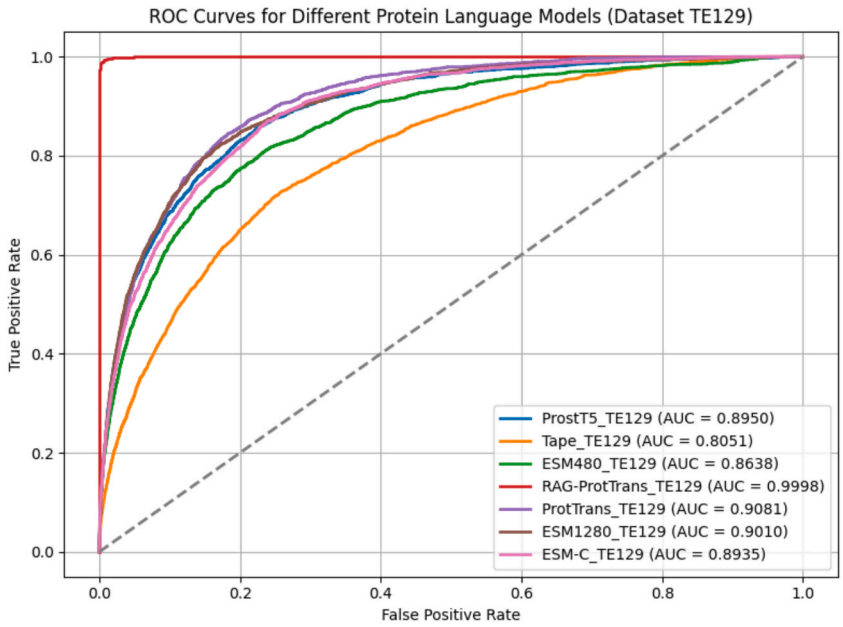



**Fig. 3.** ROC-AUC comparisons of RAG-ProtTrans with different pre-trained language model embeddings on the test set (TE129).

a precision of 0.9768, F1-scores of 0.9217, MCCs of 0.9163, and AUCs of 0.9896. ProtTrans embeddings, on the other hand, showed a significant reduction in recall (0.2808) and F1-score (0.3820), suggesting poor sensitivity to binding residues, while other models, such as ESM-1280, ESM-C, and ProstT5, improved recall slightly but failed to achieve a balanced trade-off. Fig. 4 illustrates the ROC-AUC of these comparisons. The results confirm that the RAG-based enhancement substantially boosts the model's ability to detect true binding sites while maintaining high precision, particularly in small and imbalanced datasets like TE46.

### 4.3. Comparison of classification models

Table 5 compares traditional machine learning models (KNN, RF), the previously developed mCNN architecture, and the proposed msCNN architecture across three test sets (TE181, TE129, and TE46). As a result of the evaluation, msCNN consistently outperformed mCNN and machine learning baselines. msCNN performed exceptionally well on the TE181 and TE129 datasets, with F1-scores of 0.9123 and 0.9514, respectively, and MCC values of 0.9114 and 0.9492. In the TE46 dataset, which has a smaller size and greater imbalance, the msCNN model performed exceptionally well with an F1-score of 0.9217 and an MCC of 0.9163, significantly exceeding those of the mCNN model at 0.8422 (F1) and 0.8313 (MCC). Conversely, KNN and RF had poor recall and MCC across all datasets, which indicates a limited ability to identify true binding residues. msCNN's superiority is further supported by the ROC-AUC curves illustrated in Figs. 5, 6, and 7, which show that msCNN consistently achieves the highest AUC values near perfect, demonstrating its ability to discriminate reliably and to be generalizable across

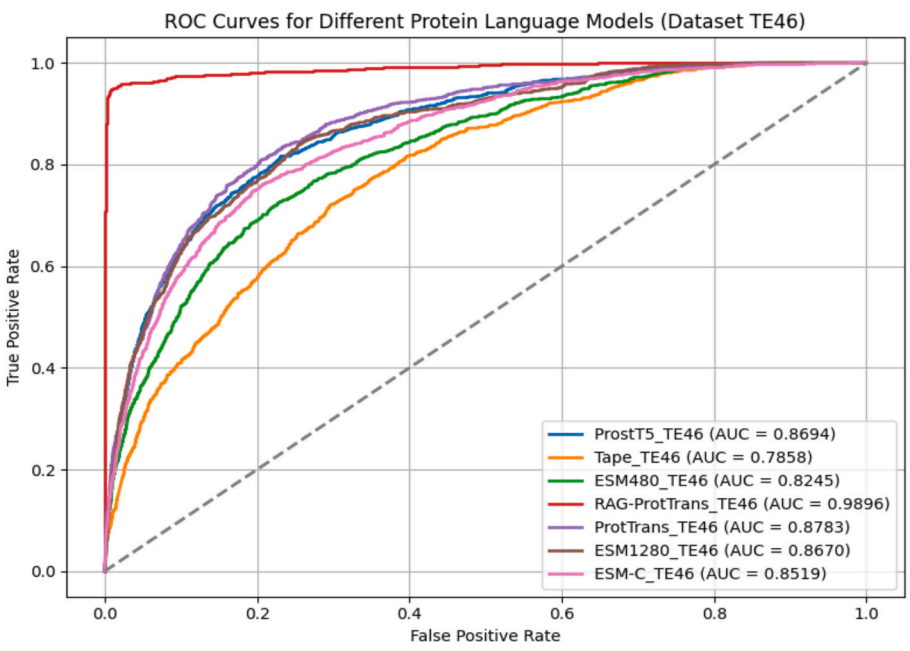



**Fig. 4.** ROC-AUC comparisons of RAG-ProtTrans with different pre-trained language model embeddings on the test set (TE46).

**Table 5**
Comparative analysis of msCNN performance with various classification methodologies with the proposed RAG-ProtTrans embeddings.

| Dataset | Model | Spe | Rec | Pre | F1 | MCC | AUC |
|---|---|---|---|---|---|---|---|
| TE181 | KNN | 0.4750 | 0.8655 | 0.0680 | 0.1261 | 0.1377 | 0.7226 |
| | RF | 0.9999 | 0.0217 | 0.9452 | 0.0424 | 0.1398 | 0.9098 |
| | mCNN | 0.9799 | 0.9855 | 0.6842 | 0.8077 | 0.8123 | 0.9988 |
| | **msCNN** | **0.9918** | **0.9934** | **0.8434** | **0.9123** | **0.9114** | **0.9998** |
| TE129 | KNN | 0.5255 | 0.8991 | 0.1074 | 0.1919 | 0.2012 | 0.7799 |
| | RF | 1.0000 | 0.0326 | 0.9865 | 0.0631 | 0.1739 | 0.9335 |
| | mCNN | 0.9903 | 0.9857 | 0.8659 | 0.9857 | 0.9188 | 0.9984 |
| | **msCNN** | **0.9938** | **0.9955** | **0.9109** | **0.9514** | **0.9492** | **0.9998** |
| TE46 | KNN | 0.8575 | 0.3306 | 0.1843 | 0.2366 | 0.1462 | 0.5941 |
| | RF | 0.9603 | 0.1523 | 0.2722 | 0.1953 | 0.1475 | 0.7306 |
| | mCNN | 0.9704 | 0.9482 | 0.7575 | 0.8422 | 0.8313 | 0. 9850 |
| | **msCNN** | **0.9980** | **0.9523** | **0.9768** | **0.9217** | **0.9163** | **0.9896** |

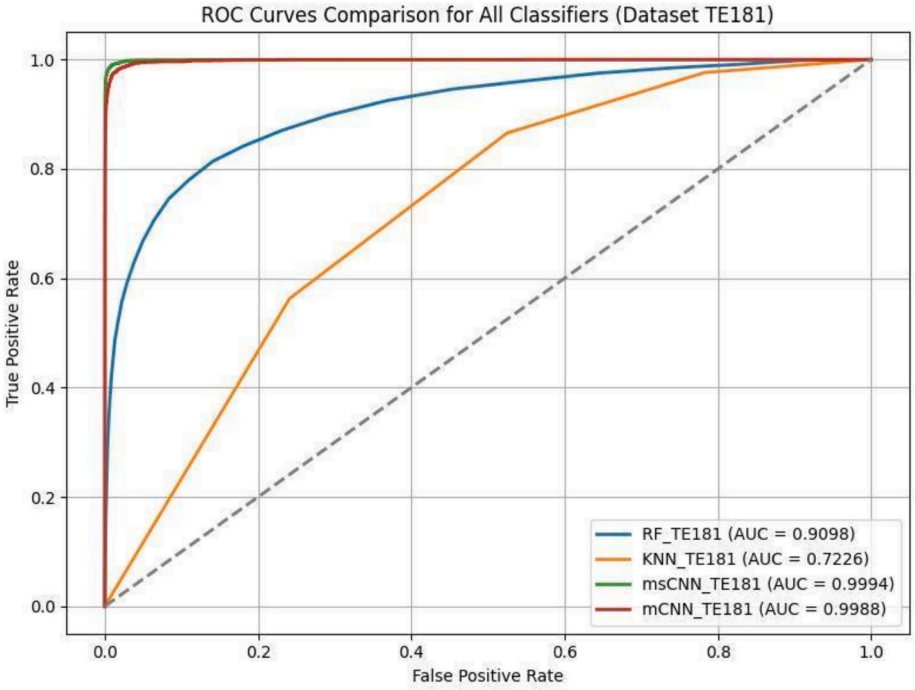



**Fig. 5.** The ROC-AUC comparison of the msCNN model with the machine learning classifier and the deep learning model mCNN on the test set (TE181).

a wide range of test conditions. On all datasets, the RAG-msCNN strategy consistently outperformed both classical and deep learning-based sequence predictors, with an improvement across all evaluation metrics.

### *4.4. Proposed model comparison against existing methods on a benchmark dataset*

We evaluated the performance of the RAG-msCNN framework using three standard benchmark datasets: TE46, TE181, and TE129, comparing it against several existing sequence-based protein-DNA binding residue predictors, as shown in Table 6. PDNAPred [42] was the primary source of performance metrics for comparison models, including DRNAPred [43], DNAPred [14], SVMnuc [44], NCBRPred [45], DBPred [40], and CLAPE-DB [46], and was adopted from the PDNAPred study, which compiles results from CLAPE-DB. According to the PDNAPred study, ULDNA [47] results were obtained from PDNAPred as well but computed using the publicly available standalone implementation of ULDNA. This ensures that the comparison values presented in our evaluation are the same as those reported in PDNAPred.

On the TE46 dataset, we found excellent performance with a specificity of 0.9980, a recall of 0.9523, a precision of 0.9768, an F1-score of 0.9217, an MCC of 0.9163, and an AUC of 0.9896. On the same dataset, PDNAPred, the best-performing existing method, achieved F1-scores of 0.539 and MCCs of 0.493. RAG-msCNN shows excellent balance between sensitivity and precision in the differentiation of binding from non-binding residues.

On the TE181 dataset, RAG-msCNN outperformed all baseline methods with a specificity of 0.9918, a recall of 0.9934, a precision of 0.8434, an F1-score of 0.9123, an MCC of 0.9114, and an AUC of 0.9998. Based on the comparison with PDNAPred, RAG-msCNN achieved an F1-score of 0.386 and an MCC of 0.365. The ability of RAG-msCNN to maintain high performance across a larger and more diverse dataset demonstrates its generalization ability.

On the TE129 dataset, the RAG-msCNN achieved a specificity of 0.9938, a recall of 0.9955, a precision of 0.9109, an F1-score of 0.9514, an MCC of 0.9492, and an AUC of 0.9999. Compared to existing methods, PDNAPred, which showed the best performance on this set, reported an F1-score of only 0.523 and an MCC of 0.494. We have demonstrated the superiority of our model, particularly with respect to the balanced classification power (MCC) and the overall accuracy (F1-

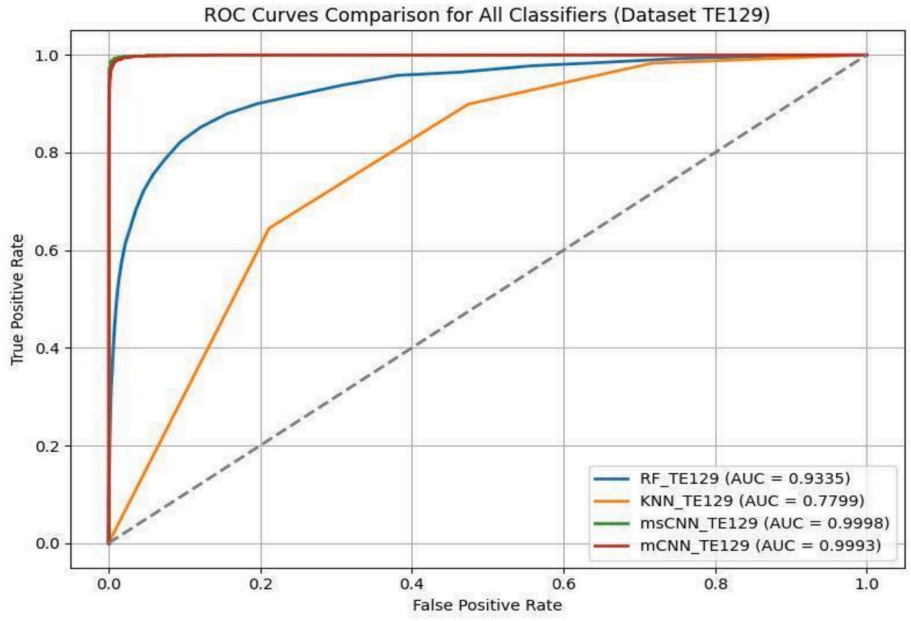



**Fig. 6.** The ROC-AUC comparison of the msCNN model with the machine learning classifier and the deep learning model mCNN on the test set (TE129).

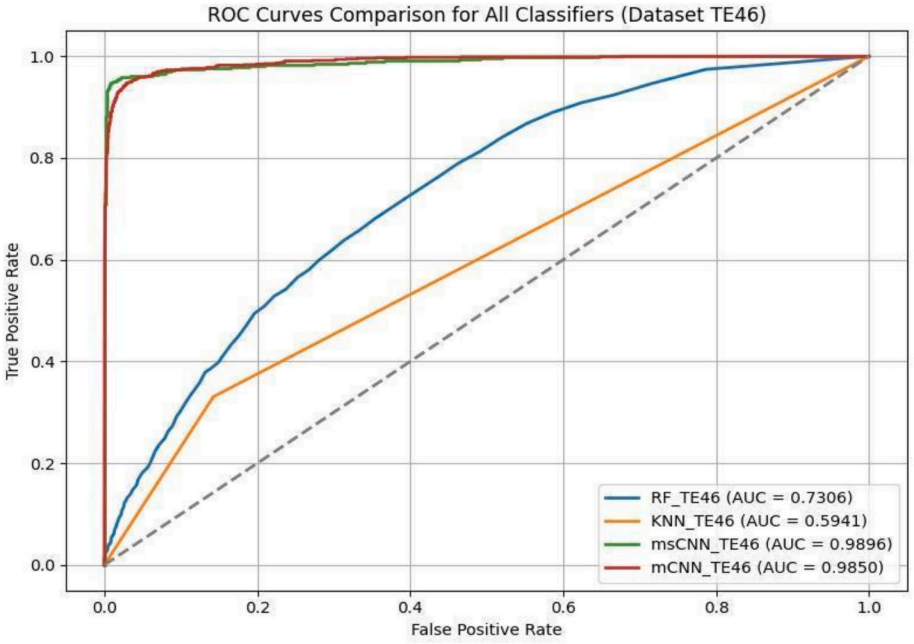



**Fig. 7.** The ROC-AUC comparison of the msCNN model with the machine learning classifier and the deep learning model mCNN on the test set (TE46).

score and AUC) of residue-level discrimination.

### *4.5. Model interpretation using t-SNE visualizations and SHAP-based feature attributes*

The RAG-msCNN DNA-binding site prediction model was analyzed comprehensively using t-distributed Stochastic Neighbor Embedding (t-SNE) and SHapley Additive Explanations (SHAP) visualizations to understand how it improves DNA-binding site predictions. Both analyses provide insight into feature contribution at the residue level (SHAP) as well as representation learning quality (t-SNE).

In Fig. 8, t-SNE visualizations compare ProtTrans alone and ProtTrans-RAG integrated embeddings, where each point represents a residue, colored by class as DNA-binding yellow and non-DNA-binding purple. In Figure (A), there is some separation between classes, and the two distributions are highly overlapping. This indicates that the embeddings do capture class-specific signals, but not with a high degree of discrimination. A diffuse boundary with mixed clusters separates DNA-binding residues from non-binding residues. In Figure (B), RAG-ProtTrans is shown, resulting in significantly improved separation. A more cohesive cluster of DNA-binding residues results in less overlap with non-binding residues, showing that RAG enhances context-aware learning and captures physicochemical and structural characteristics better. This enhanced separability reflects the RAG fusion's enhancement of contextual representation, which allows residues linked with DNA-binding sites to form more coherent and physiologically

**Table 6**
Performance comparison of RAG-msCNN with other sequence-based predictors.

| Dataset | Model | Spe | Rec | Pre | F1 | MCC | AUC |
|---|---|---|---|---|---|---|---|
| TE46 | DRNAPred | 0.692 | 0.677 | 0.185 | 0.291 | 0.226 | 0.775 |
| | DNAPred | 0.655 | 0.671 | 0.157 | 0.254 | 0.194 | 0.730 |
| | SVMnuc | 0.666 | 0.668 | 0.154 | 0.250 | 0.192 | 0.715 |
| | NCBRPred | 0.674 | 0.677 | 0.165 | 0.265 | 0.207 | 0.713 |
| | DBPred | 0.784 | 0.708 | 0.243 | 0.362 | 0.320 | 0.794 |
| | CLAPE-DB | 0.835 | 0.747 | 0.306 | 0.434 | 0.401 | 0.871 |
| | ULDNA [47] | 0.696 | 0.800 | 0.204 | 0.325 | 0.296 | 0.831 |
| | PDNAPred | 0.949 | 0.561 | 0.519 | 0.539 | 0.493 | 0.897 |
| | **RAG-msCNN** | **0.9980** | **0.9523** | **0.9768** | **0.9217** | **0.9163** | **0.9896** |
| TE181 | DNAPred | 0.948 | 0.334 | 0.223 | 0.267 | 0.233 | 0.802 |
| | SVMnuc | 0.960 | 0.289 | 0.242 | 0.263 | 0.229 | 0.803 |
| | NCBRPred | 0.964 | 0.259 | 0.241 | 0.250 | 0.215 | 0.771 |
| | CLAPE-DB | 0.931 | 0.413 | 0.212 | 0.280 | 0.252 | 0.824 |
| | ULDNAb | 0.917 | 0.585 | 0.238 | 0.339 | 0.331 | 0.851 |
| | PDNAPred | 0.949 | 0.512 | 0.309 | 0.386 | 0.364 | 0.896 |
| | **RAG-msCNN** | **0.9918** | **0.9934** | **0.8434** | **0.9123** | **0.9114** | **0.9998** |
| TE129 | DRNAPred | 0.937 | 0.233 | 0.190 | 0.210 | 0.155 | 0.693 |
| | DNAPred | 0.954 | 0.396 | 0.353 | 0.373 | 0.332 | 0.845 |
| | SVMnuc | 0.966 | 0.316 | 0.371 | 0.341 | 0.304 | 0.812 |
| | NCBRPred | 0.969 | 0.312 | 0.392 | 0.347 | 0.313 | 0.823 |
| | CLAPE-DB | 0.955 | 0.464 | 0.396 | 0.427 | 0.389 | 0.881 |
| | ULDNAb | 0.911 | 0.725 | 0.340 | 0.463 | 0.452 | 0.893 |
| | PDNAPred | 0.957 | 0.595 | 0.466 | 0.523 | 0.494 | 0.923 |
| | **RAG-msCNN** | **0.9938** | **0.9955** | **0.9109** | **0.9514** | **0.9492** | **0.9999** |

meaningful clusters, enhancing model interpretability and prediction accuracy.

In Fig. 9, the SHAP summary plots show how individual amino acid positions (features) contribute to the prediction output for both models. Each dot represents a SHAP value for a particular feature in one sample, with red representing high and blue representing low. ProtTrans in Figure (A) shows that a few features (e.g., F402, F523, F884) dominate the decision process. In contrast, many other features show near-zero contributions, suggesting sparse decision-making and possible over-reliance on a narrow set of positions. In Figure (B), RAG-ProtTrans illustrates the distribution of feature attributes across multiple residues. Several features (F814, F402, F964) still play a significant role, but several additional positions also contribute significantly. The balanced SHAP distribution enables a more context-aware model based on sequence and structural information. Features with high SHAP values correspond to residues that are commonly involved in electrostatic or hydrogen bond interactions with DNA, showing that the model's learned features are associated with known binding mechanisms. RAG integration not only increases overall performance but also improves biological plausibility by distributing importance over a wider set of residues, possibly aligning with actual DNA-contacting sites.

### *4.6. Evaluation of the proposed architecture on RNA dataset (RNA117 test Set)*

A comparative analysis was conducted using the RNA117 test set against a wide range of state-of-the-art RNA-binding site predictors to evaluate the performance and generalization ability of our proposed model. It includes RNABindPlus [15], SVMnuc [36], CLAPE-RB [38], COACH-D [48], NucBind [44], aaRNA [16], NucleicNet [17], two versions of GraphBind [19], and the recent PDNAPred [42] framework. According to PDNAPred, two performance reports were provided for GraphBind for the same dataset, one from the original publication and one from the authors' reimplementation in Table 7. Our model achieved a recall of 0.9717, a precision of 0.9664, an F1-score of 0.9690, an MCC of 0.9431, and an AUC of 0.9963, which represents a significant improvement over PDNAPred's F1-score of 0.3215 and MCC of 0.274, as well as GraphBind's best versions of 0.3358 and 0.322, and the AUC of our model is nearly accurate for distinguishing RNA-binding residues from non-binding residues. Through the integration of pretrained ProtTrans embedding and the RAG strategy, a multiscale separable CNN architecture can capture long-range interactions. RAG-msCNN operates entirely on sequence-based information and exhibits robust generalization, unlike several other methods that rely on structural data.

## 5. Discussion

Accurate identification of protein-DNA and RNA binding sites is essential for understanding gene regulation, genome stability, and post-transcriptional control mechanisms. Compared to existing machine learning approaches and cutting-edge predictors like DBPred and CLAPE-DB, our RAG-msCNN outperformed all benchmark datasets. The significant increases in AUC and F1 scores show that combining pre-trained embeddings with RAG integration and multiple-scale separable convolutional architectures improves the ability to distinguish protein-DNA binding sites.

The RAG module uses a weighted attention approach to get contextually related residues and combine their embeddings with the target residue representation. The embedding obtained helps to refine the representation of the target residue by emphasizing biologically important contextual information. The retrieval weights help us understand how contextual information helps us recognize binding sites by showing how comparable residues affect the prediction process.

Our comparison evaluation ensures transparency, credibility, and reproducibility by clearly indicating the sources of the comparison values. In addition to improving computational tools for annotation of biomacromolecules with significant biological implications, a model's prediction ability may also allow for the identification of novel regulatory elements, the guidance of protein engineering efforts, and the development of therapeutic interventions targeting protein-nucleic acid interactions.

## 6. Conclusion

Our study presents a novel deep learning framework, RAG-msCNN, to identify nucleic acid-binding residues accurately utilizing Retrieval-Augmented Generation (RAG) and pre-trained protein language model embeddings, as well as a separable convolutional neural network (msCNN) that effectively captures residue-level spatial patterns and hierarchical features. This unified architecture has been applied to the prediction of DNA and RNA-binding sites, and it consistently outperforms previous methods in terms of recall, precision, F1, MCC, and

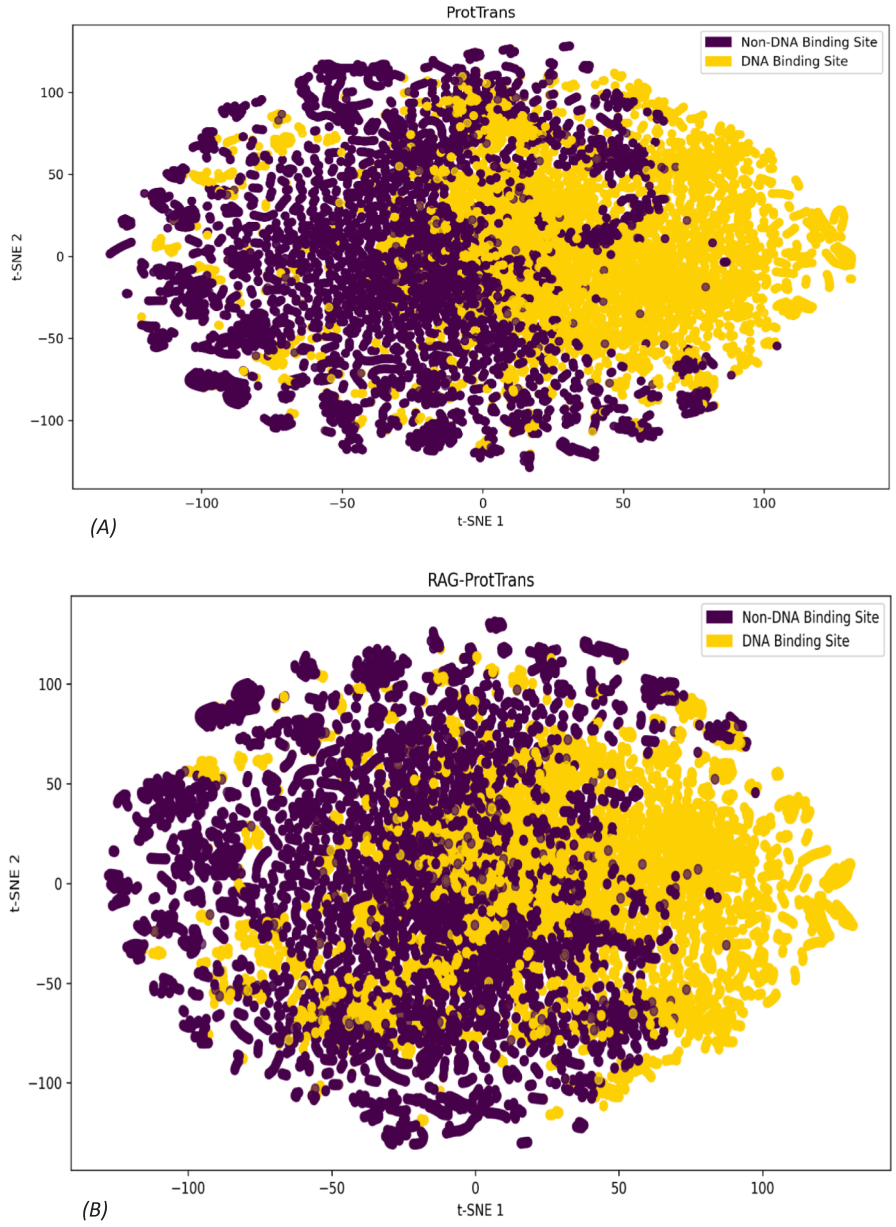



**Fig. 8.** T-sne visualization of residue-level embeddings for dna-binding site prediction: *(A)* ProtTrans embeddings, *(B)* RAG-ProtTrans embeddings.

AUC.

The results of this study demonstrate that our RAG-enhanced strategy captures long-range residue dependencies, particularly on challenging benchmark datasets such as DNA129 and RNA117. In comparison to competitive predictors like PDNAPred, GraphBind, and NucleicNet, our proposed RAG-msCNN demonstrated superior generalization and sensitivity to functional binding residues.

In addition to numerical performance, we emphasize interpretability through t-SNE and SHAP visualizations. RAG integration improved feature discrimination by separating binding and non-binding residues in the latent embedding space. Based on the analysis of SHAP values, it was possible to determine the key residues involved in the decision-making process mode. This study develops a biologically sound and technologically advanced method for predicting high-confidence protein-nucleic acid interactions. Given the significance of such interactions in gene regulation, this framework is expected to help with large-scale functional annotations and experimental studies in structural and molecular biology.

Future research will focus on developing the proposed RAG-msCNN framework to include dynamic and temporal modeling techniques, such as nonlinear autoregressive exogenous (NARX) networks, to better reflect sequential dependencies and residue interaction patterns in protein structures. Furthermore, fractional or adaptive intelligent architectures could be investigated to improve biological interpretability and prediction robustness, as suggested by current research on intelligent adaptive Bayesian and exogenous network models.

## Ethical approval

This study utilized publicly available protein sequence. No experiments involving human participants or animals were conducted.

## CRediT authorship contribution statement

**Muhammad Shahid Malik:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal

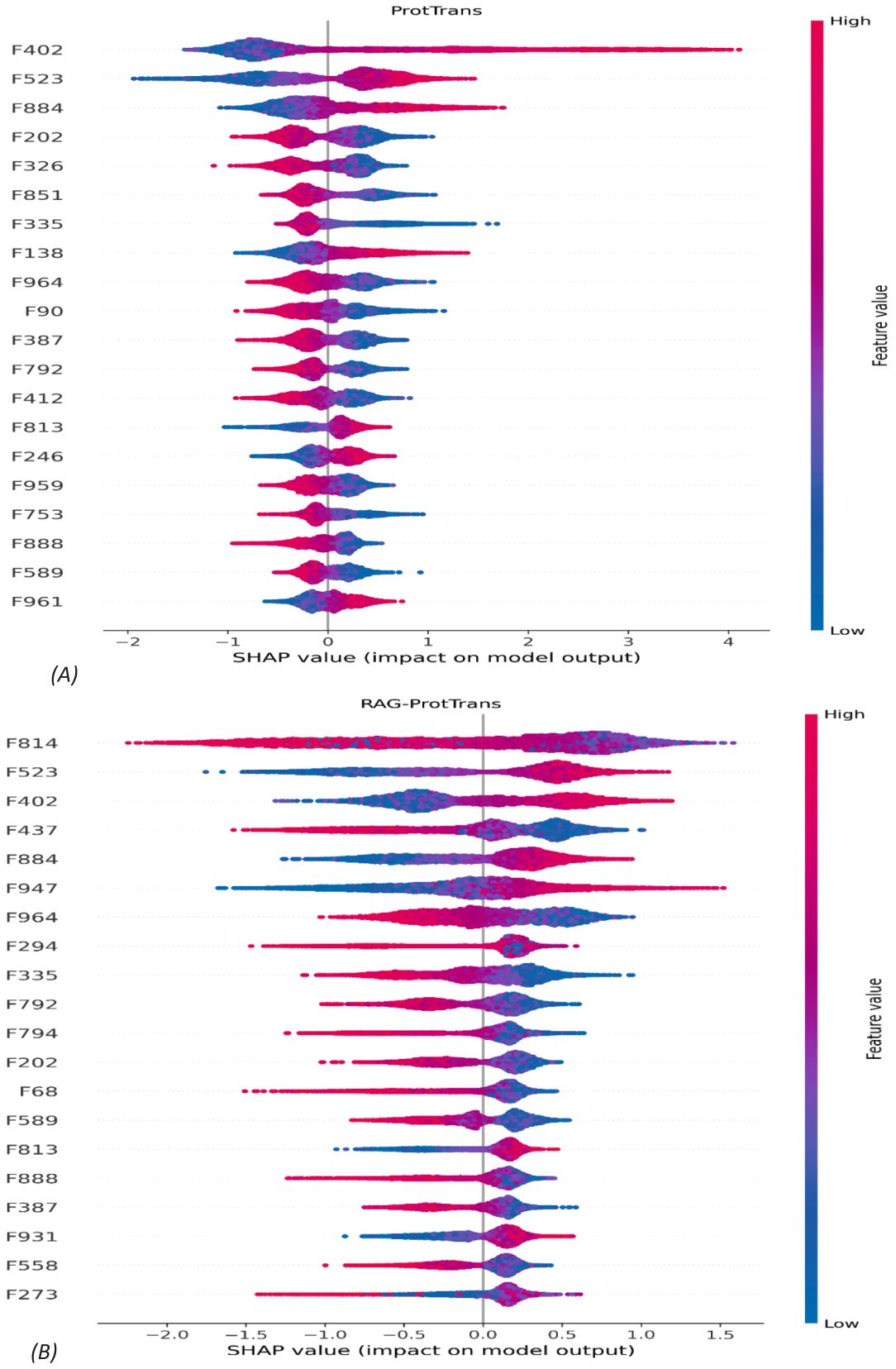



**Fig. 9.** SHAP value interpretation of model predictions: *(A)* ProtTrans embeddings; *(B)* RAG-ProtTrans embeddings.

**Table 7**
Performance comparison of PDNAPred and Other Methods on the RNA117 Test Set.

| Models | Rec | Pre | F1 | MCC | AUC |
|---|---|---|---|---|---|
| RNABindPlus | 0.273 | 0.227 | 0.248 | 0.202 | 0.717 |
| SVMnucc | 0.231 | 0.240 | 0.235 | 0.192 | 0.729 |
| CLAPE-RB | 0.467 | 0.201 | 0.281 | 0.240 | 0.800 |
| COACH-D | 0.221 | 0.252 | 0.235 | 0.195 | 0.663 |
| NucBind | 0.231 | 0.235 | 0.233 | 0.189 | 0.715 |
| aaRNA | 0.484 | 0.166 | 0.237 | 0.214 | 0.771 |
| NucleicNet | 0.371 | 0.201 | 0.261 | 0.216 | 0.788 |
| GraphBind | 0.303 | 0.171 | 0.218 | 0.168 | 0.718 |
| GraphBind2 | 0.463 | 0.294 | 0.358 | 0.322 | 0.854 |
| PDNAPred | 0.335 | 0.298 | 0.315 | 0.274 | 0.829 |
| **RAG-msCNN** | **0.9717** | **0.9664** | **0.9690** | **0.9431** | **0.9963** |

analysis, Conceptualization. **Muhammad Hussain:** Writing – original draft, Visualization, Validation, Software, Investigation, Data curation. **Quang-Thai Ho:** Supervision, Project administration, Data curation, Conceptualization. **Van-The Le:** Validation, Software, Data curation. **Yu-Yen Ou:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Our code and dataset can be found on GitHub at: https://github.com/mhb1717/RAG_DNA_RNA.

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